Pranav Lakhe

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SUMMARY

3rd-year B.Tech CS & IT student building AI/ML systems, computer-vision pipelines, and agentic workflows. Published patent inventor, MARL research author, and Smart India Hackathon '26 finalist.

EDUCATION

B.Tech, Computer Science & IT — Symbiosis Institute of Technology, Nagpur (2024–2028) | CGPA: 7.53/10

SKILLS

- **Languages:** Python, JavaScript, TypeScript, C, C++, Java, Bash
- **AI/ML:** PyTorch, TensorFlow, scikit-learn, YOLOv8/v11, ByteTrack, DeepSORT, OpenCV, MARL, Ensemble Learning, Langflow, watsonx.ai, Claude API
- **Web/Data/DevOps:** React, FastAPI, Node.js, Streamlit, Gradio, PostgreSQL, Redis, Docker, Git, REST, RTSP/ONVIF

PROJECTS

- **AI Border Surveillance Video Analytics** (SIH 2026 Finalist, MHA/SSB) — Converts legacy CCTV into an intrusion-detection network using YOLOv11, ByteTrack, virtual fences, and ANPR with offline-first edge processing (FastAPI, React, Docker).
- **SHREEDHA Hospital RoI Dashboard** — Healthcare financial-intelligence platform with revenue-leakage radar, anomaly detection, equipment-ROI simulator, and an LLM financial copilot (FastAPI, PostgreSQL, Claude API).
- **SaafReport, AI Medical Report Summarizer** — Agent that turns clinical PDFs into calm, categorized summaries with PDF/JSON export (Claude, pdfplumber, Gradio).
- **California Housing Price Predictor** — Compared linear and tree models, tuned with GridSearchCV to $R^2 = 0.8456$; deployed on Streamlit.
- **Arduino Ultrasonic Radar System** (coursework prototype) — Servo-swept HC-SR04 sensor (0°–180°, 2–400 cm range) streaming distance data over 9600-baud UART to a Processing IDE radar GUI, with 16x2 LCD and LED indicators.
- **MemoryReview** (Nexus Hackathon) — AI code reviewer with persistent memory across reviews.

PATENT & RESEARCH

- **Published Patent (Govt. of India), App. No. 202521125542 A** — First inventor of an ensemble ML system (voting regression + gradient boosting) predicting student performance with $R^2 > 0.89$.
- **MARL for Autonomous Urban Fleets** — Authored a 12-page study on decentralized multi-agent RL: shared rewards, coordination, deadlock avoidance.
- **ACEx Multi-Agent AI Coding Assistant** — Authored a 7-page IEEE-style paper formulating Assist, Explain, and Algo-Generator agents, synthesizing 16+ citations on developer ergonomics, cognitive load, and enterprise deployment.

EXPERIENCE

- **IBM SkillsBuild × AICTE, Agentic AI Intern (2026)** — Built an autonomous Interview Trainer Agent with Langflow and IBM Granite on watsonx.ai, with conversational guardrails, adaptive questioning, and skill-assessment rubrics for technical screening.
- **Maincrafts Technology, ML Pipeline Intern (3 months)** — Built end-to-end regression pipelines covering preprocessing, feature engineering, and GridSearchCV tuning (validation $R^2 = 0.8456$); delivered Streamlit apps and IEEE-format reports.
- **VOIS × AICTE, Cybersecurity Intern (Mar–Apr 2026)** — Built an asynchronous, multi-threaded Python network port scanner GUI (Tkinter, sockets); identified open DNS services on Port 53 and assessed amplification and zone-transfer risks.
- **ICSIT 2026, Technical Volunteer & Hybrid Ops Lead** — Ran session management, AV/stream coordination, and hybrid operations for the 2nd International Conference on Sustainability, Innovation & Technology (Sep 26–27, 2026) at SIT Nagpur.

ACHIEVEMENTS

- Smart India Hackathon '26 Finalist (SIH26187, Team Hyrex Coders) • Published Govt. of India Patent (Journal No. 1/2026) • MARL research author

CERTIFICATIONS

Red Hat System Administration • Microsoft Data Analytics • Nexus Hackathon (MemoryReview, AI code reviewer) • IBM SkillsBuild × AICTE • VOIS × AICTE Cybersecurity